SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT1 — CHART CONSTRUCTION TEST

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Construction (BL3)	**Assessment:**	Assessment Tool 1 – Chart Construction
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO2 Statement:	*Apply appropriate chart construction techniques — correct chart type, axes, scales, and calculations — to accurately represent given data.(BL3)*		
Student Name:	K. Amrutha Sandhya	**Reg. No.:**	192524270
Section / Batch:	AIDS	**Date:**	30-7-2026

Code of Conduct

I K. Amrutha Sandhya(192524270) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., percentages, angles, bin ranges, five-number summaries) as part of your answer.
- Every chart must include a title, correctly labeled axes, and a legend where applicable.

Annexure A – CHART CONSTRUCTION

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Chart Construction Test

CO2 · BT3: Apply ·

This rubric is used to assess student responses to the CO2-AT1 Chart Construction Test, in which students are given a small dataset and must construct an accurate, correctly labeled chart of a specified type.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of the correct chart type, and the axes, scale, and elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type, axes, or scale.	Poor understanding of core construction concepts; selects a chart type or elements fundamentally unsuited to the data.
Explanation	**25%**	Shows detailed, clearly worked calculations (e.g., angles, bin ranges, five-number summary) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	**20%**	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	**15%**	The chart is neatly and logically	The chart is well organized, with only minor issues	Basic construction; required elements	Poorly constructed; the chart is missing

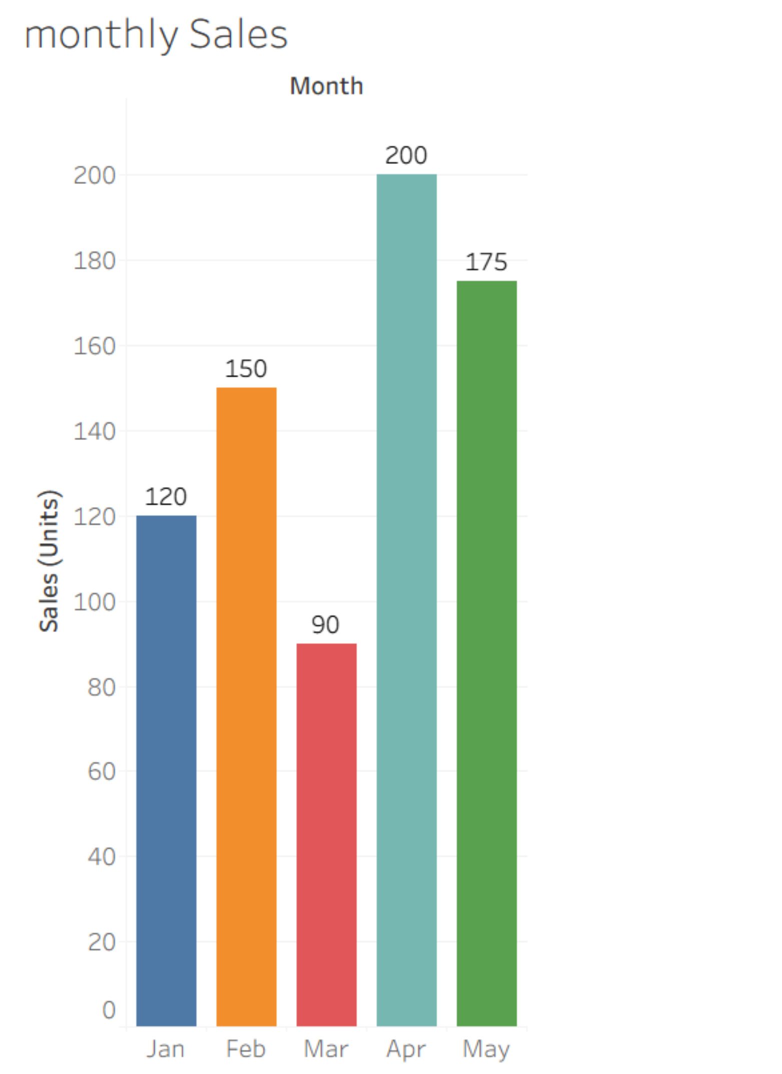
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		constructed, with a title, correctly labeled axes, a legend where applicable, and elements in a clear order.	in labeling, legend, or layout.	(title, axes, legend) are present but poorly arranged or partly missing.	key elements such as a title, axis labels, or a legend.
Accuracy	**10%**	The chart is completely accurate — all values, proportions, scales, and calculations correctly match the given data.	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

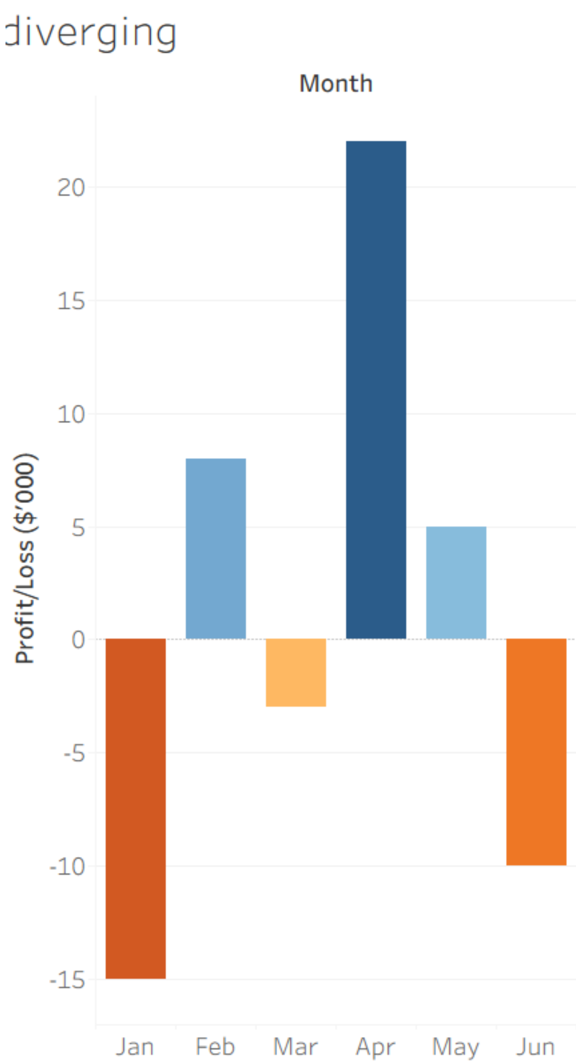
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

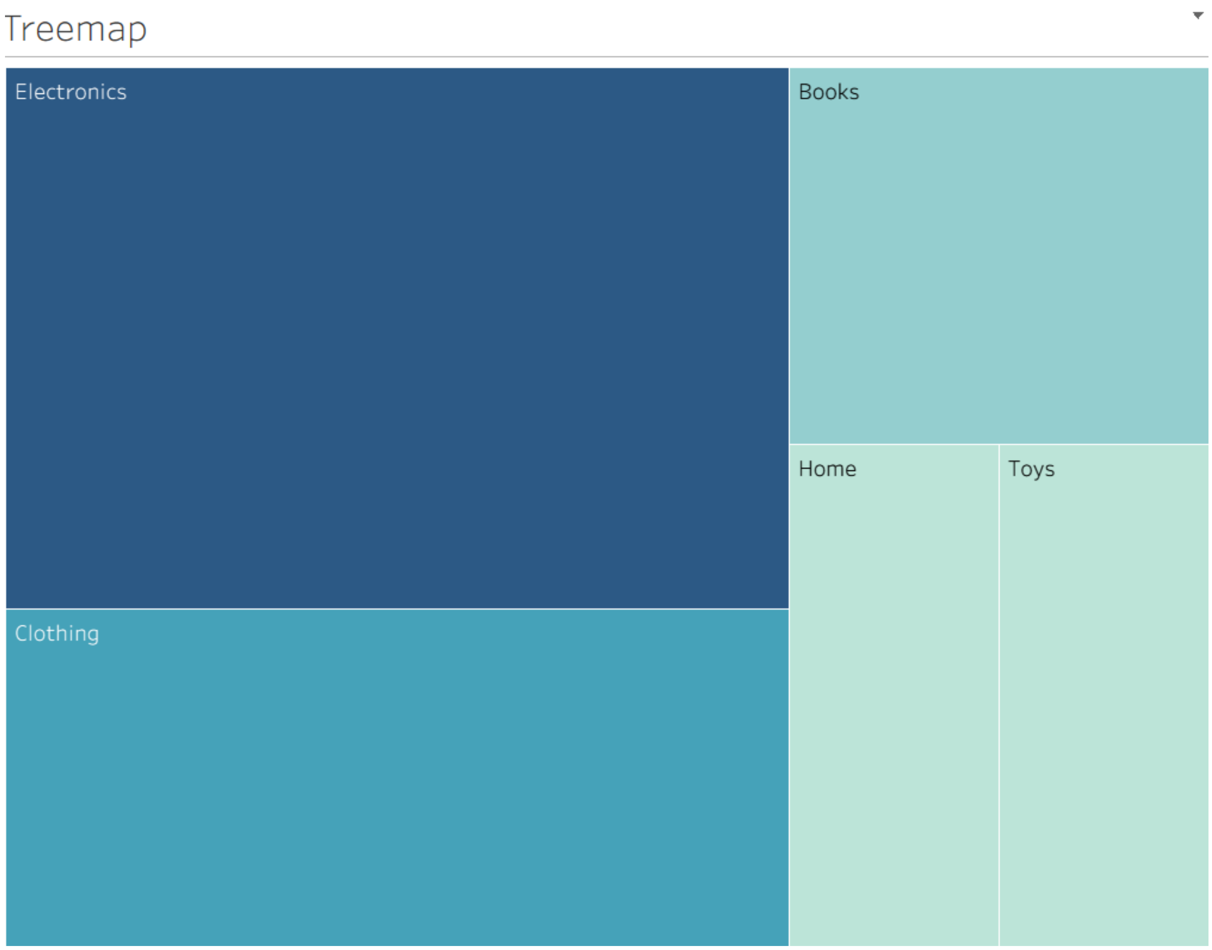
Problem 1 — Bar Chart — Basic Construction
Data: Monthly sales (units): Jan = 120, Feb = 150, Mar = 90, Apr = 200, May = 175.
Task: Construct a bar chart to represent monthly sales. Label both axes, include a chart title, and keep
the months in chronological order.



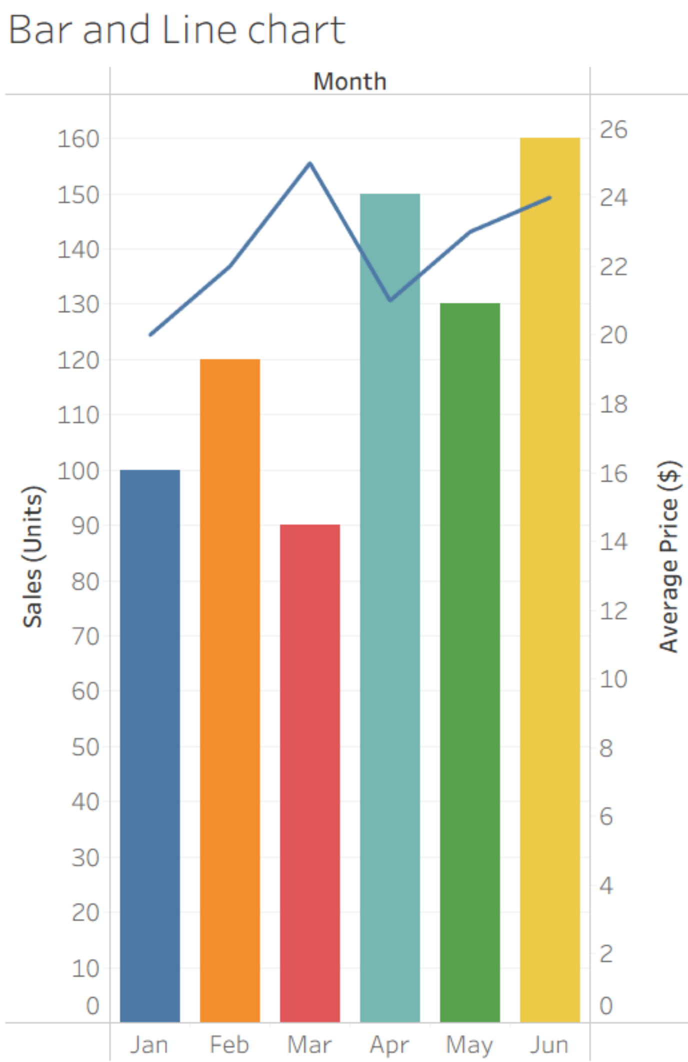
Problem 22 — Diverging Color Scale — Profit/Loss
Data: Monthly profit/loss ($'000): Jan=−15, Feb=+8, Mar=−3, Apr=+22, May=+5, Jun=−10.
Task: Construct a bar chart using a diverging color scale (e.g., one color for loss, another for profit) to
represent this data, clearly marking the zero baseline.



Problem 23 — Treemap — Proportional Area
Data: Revenue by product category: Electronics=$400K, Clothing=$250K, Books=$150K, Toys=$100K,
Home=$100K.
Task: Construct a treemap where each rectangle's area is proportional to that category's share of total
revenue. Show your calculation of each category's percentage share.



Problem 24 — Combination Chart — Bar + Line
Data: Monthly sales (units) and average price ($): Jan=(100,20), Feb=(120,22), Mar=(90,25),
Apr=(150,21), May=(130,23), Jun=(160,24).
Task: Construct a chart combining a bar chart (units sold) with a line chart (average price) across the six
months. Explain why you would avoid using a dual y-axis and how you would represent both variables
clearly instead.



Problem 25 — Small Multiples Construction
Data: Weekly sales (units) for 4 regions over 4 weeks: Region A: 40, 55, 50, 65
Region B: 30, 35, 45, 40
Region C: 60, 58, 70, 75 Region D: 20, 25, 22, 30
Task: Construct a set of four small-multiple line charts, one per region (rather than one overlaid chart),
and briefly justify this design choice over a single combined chart.